A reusable Bayesian IRT workflow for small-sample instrument validation: A tutorial with a case study in teacher belief measurement

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Abstract

Bayesian Item Response Theory (IRT) offers distinct advantages for fitting complex measurement models under sample-size constraints, yet practical guidance on implementing complete Bayesian IRT workflows for applied researchers remains limited. This tutorial presents a reusable workflow for validating multidimensional instruments with limited samples using the R package brms. The workflow guides researchers through five integrated steps: (1) preprocessing decisions for sparse ordinal responses, (2) comparing competing dimensional structures including bi-factor models, (3) using discrimination parameters as diagnostic tools for identifying threshold anomalies and misfitting items, (4) selecting among ordinal link functions using cross-validation and stability diagnostics, and (5) incorporating covariates and differential item functioning analysis. Throughout, we emphasise how the rich diagnostic information provided by Bayesian posterior distributions supports informed qualitative judgment at each decision point. The workflow is illustrated through the validation of a 24-item instrument measuring Chinese secondary mathematics teachers' beliefs about equity (n = 155). The iterative process identified misfitting items, resolved threshold anomalies, and ultimately supported a bi-factor graded response model with a general inclusive belief factor and three domain-specific components. All analysis code and data are available in a companion repository.

Keywords: Bayesian item response theory; multidimensional IRT; instrument validation; Bayesian workflow; brms

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Introduction

Bayesian IRT: A brief overview

Diaggnostic toolkit

Case study: Data and setup

Case Study: Data and Setup

Dataset

The case study uses questionnaire data from secondary mathematics teachers (*n* = 155) in public schools across three urban districts in a city in Zhejiang province, China. The three districts are coded 1, 2, and 3 in descending order of socioeconomic status. Three demographic variables were retained for subsequent differential item functioning analysis: teacher gender (male/female), school district (1/2/3), and the year group taught (7/8/9).

Instrument

The instrument comprises 24 items measuring the degree to which teachers hold inclusive, equity-oriented beliefs about mathematics education, as opposed to more traditional or exclusionary orientations. It was adapted from the Teacher Education and Development Study in Mathematics (TEDS-M; Tatto, 2013) with cultural modifications for the Chinese context (e.g., replacing references to "hands-on experiences" with locally recognisable pedagogical reforms such as project-based learning). Items are organised into three subscales:

- *Beliefs about mathematics* (Bm, 8 items) assess whether teachers view mathematics as a fixed body of rules or as a creative, applied discipline. Items endorsing a formalist view are reverse-coded (e.g., "Mathematics is a collection of rules and procedures" [Bm_1_r, reverse-coded]).
- *Beliefs about learners* (Bl, 11 items) assess whether teachers believe mathematical ability is innate and fixed or accessible to all students (e.g., "To be good at mathematics one needs to

be talented” [Bl_7_r, reverse-coded]).

- *Beliefs about pedagogy* (Bp, 5 items) assess teachers’ receptiveness to innovative and student-centred teaching approaches (e.g., “I’m continuously seeking better ways to teach mathematics” [Bp_3]).

Teachers responded on a 5-point Likert scale from “strongly disagree” to “strongly agree.” The full item list is provided in Appendix B.

The instrument’s three-dimensional design, grounded in frameworks distinguishing what mathematics is, who can learn it, and how it should be taught (Felton-Koestler, 2017), naturally motivates competing structural hypotheses for IRT modelling.

Competing structural hypotheses

Given the instrument’s multidimensional design, we consider three structural hypotheses representing increasingly complex assumptions about the latent space Figure 1. Each hypothesis is grounded in distinct theoretical perspectives on how teacher beliefs are organised.

In the notation below, η_{pi} denotes the linear predictor for person p on item i (mapped to category probabilities by the ordinal link function), β_0 is the fixed intercept, ξ_i is item i’s difficulty, and $\mathbb{I}_{id}$ is a binary indicator assigning item i to dimension $d \in \{M, L, P\}$.

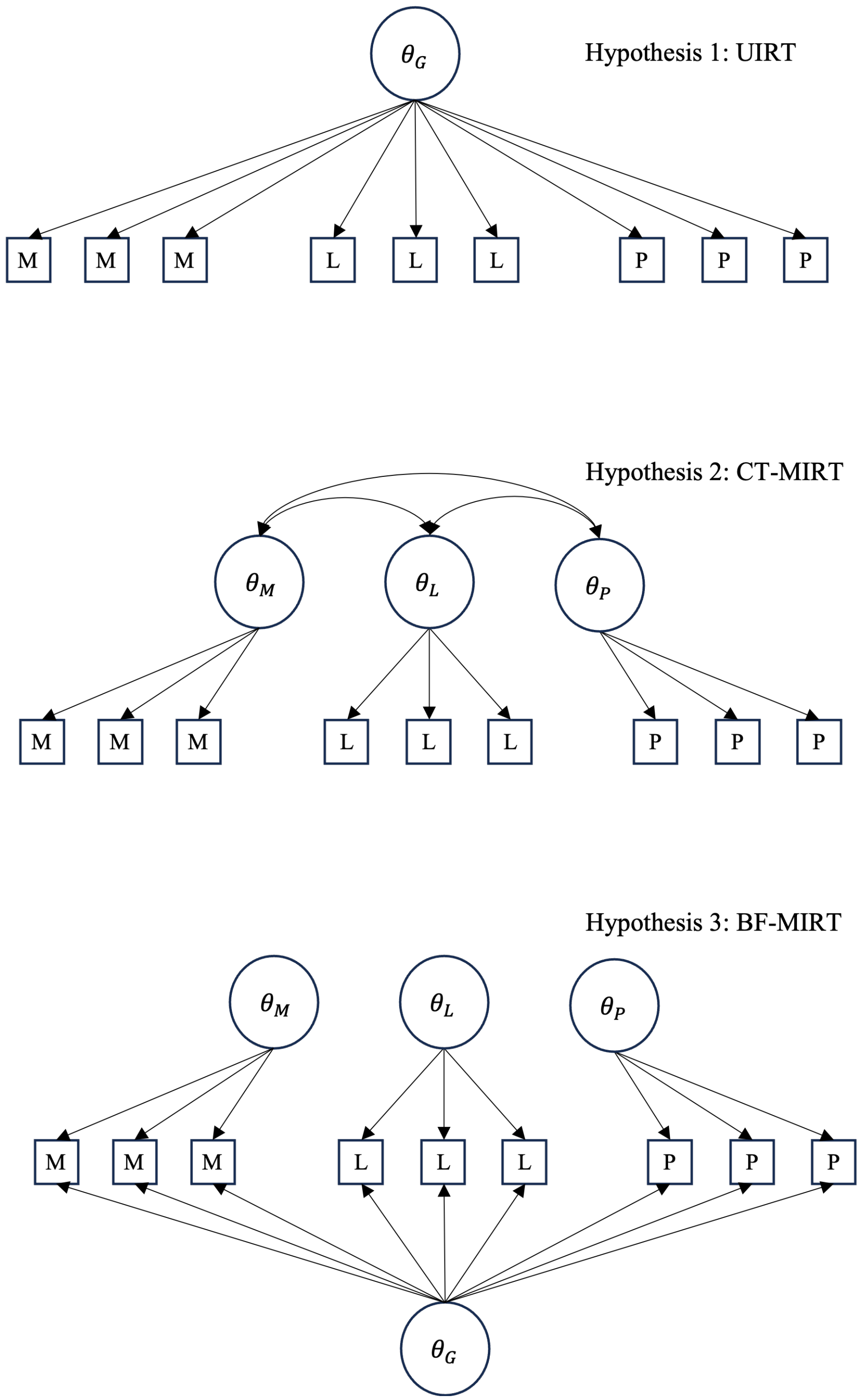
Unidimensional. The simplest assumption is that all items measure a single continuum of inclusive belief. This implies that teachers’ orientations toward mathematics, learners, and pedagogy vary in concert, so that a single latent trait θ_p suffices:

$$\eta_{pi} = \beta_0 + \xi_i + \theta_p$$

Correlated traits. According to belief cluster theory (Green, 1971), teachers’ beliefs are not monolithic but organised in clusters that may develop and function semi-independently. Applied to this instrument, the three subscales may represent distinct yet interconnected belief clusters, allowing, for instance, a teacher to hold formalist views of mathematics while still believing most students can succeed. Each person is then characterised by a vector

Figure 1

Graphical representation of three competing dimensional structures for the equity belief instrument. Top panel: unidimensional model. Middle panel: correlated-traits model. Bottom panel: bi-factor model.



$\boldsymbol{\theta}_p = (\theta_{pM}, \theta_{pL}, \theta_{pP})^\top \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ with an unrestricted covariance matrix:

$$\eta_{pi} = \beta_0 + \xi_i + \sum_{d \in \{M,L,P\}} \mathbb{I}_{id}\, \theta_{pd}$$

Bi-factor. Belief system theories (Green, 1971; Rokeach, 1968) further propose that beliefs form hierarchical networks in which some core beliefs are stable and broadly influential, while others are more peripheral and context-dependent. Statistically, this implies a bi-factor structure: a general inclusive belief factor captures the shared variance across all items, while dimension-specific residual factors capture variation not explained by the general factor. A general trait θ_p is added alongside orthogonal specific factors θ^s_{pd}:

$$\eta_{pi} = \beta_0 + \xi_i + \theta_p + \sum_{d \in \{M,L,P\}} \mathbb{I}_{id}\, \theta^s_{pd}$$

The orthogonality constraint $\mathrm{Cov}(\theta_p, \theta^s_{pd}) = 0$ is imposed here.[1] Its implementation in brms is discussed in Section 5.2. All three hypotheses can be combined with any of the ordinal link functions described in Appendix A; competing combinations are systematically compared in the workflow.

Computational setup

All models were fitted in R using the brms package (Bürkner, 2017), which interfaces with Stan (Carpenter et al., 2017) for Hamiltonian Monte Carlo sampling. Each model used 4 chains with 4,000 iterations per chain (2,000 warmup), yielding 8,000 post-warmup draws for inference. Weakly informative priors were specified following Bürkner (2021) : $\mathcal{N}(0, 3)$ for person and item random effect standard deviations, $\mathcal{N}(0, 1)$ for log-discrimination parameters, and $\mathrm{LKJ}(2)$ for correlation matrices in the correlated-traits models.[2] Convergence was assessed

[1] It is possible to relax this constraint and estimate correlations between the general and specific factors. However, doing so introduces identification difficulties, as the model must distinguish general-factor variance from correlated specific-factor variance using the same set of item responses. With small samples this also raises the risk of overfitting, where the additional covariance parameters absorb noise rather than capturing meaningful structure.

[2] The LKJ prior (Lewandowski, Kurowicka, & Joe, 2009) is a distribution over correlation matrices. The shape parameter η controls the density: $\eta = 1$ is uniform over all valid correlation matrices; $\eta > 1$ increasingly concentrates

by confirming $\hat{R} < 1.01$ for all parameters, adequate bulk and tail effective sample sizes, and no persistent divergent transitions. Additional sampler settings introduced for complex multidimensional models are discussed alongside the relevant workflow steps. Complete code is provided in the companion repository.

density near the identity matrix, favouring smaller correlations. We use $\eta = 2$ as a mild regularising choice.

References

Bürkner, P.-C. (2021). Bayesian item response modeling in R with brms and Stan. *Journal of Statistical Software*, *100*, 1–54.

Appendix A

Model specifications

This appendix provides detailed mathematical specifications of the ordinal IRT models used in this study. All models are extensions of IRT for ordered categorical responses, differing in their link functions, threshold parameterisations, and discrimination constraints. Implementation details in brms are provided in the main text and companion repository.

Ordinal Link Functions and Threshold Parameterisations

Rating Scale Model (RSM)

The RSM uses the adjacent-category logit link with shared thresholds. The probability of person p responding in category c ($c = 1, \ldots, C$) to item i is:

$$P(Y_{pi} = c) = \frac{\exp \sum_{j=1}^{c} (\theta_p + \xi_i - \tau_j)}{\sum_{r=1}^{C} \exp \sum_{j=1}^{r} (\theta_p + \xi_i - \tau_j)}$$

where θ_p is the latent trait of person p, ξ_i is the item-specific location (random intercept; more negative = harder to endorse), and τ_j is the j-th threshold parameter. A key constraint of the RSM is that thresholds τ_j are shared across all items, assuming equal category spacing across the instrument.

Generalised Rating Scale Model (GRSM)

Introducing item-specific discrimination parameters α_i to the RSM yields the GRSM:

$$P(Y_{pi} = c) = \frac{\exp \sum_{j=1}^{c} \alpha_i(\theta_p + \xi_i - \tau_j)}{\sum_{r=1}^{C} \exp \sum_{j=1}^{r} \alpha_i(\theta_p + \xi_i - \tau_j)}$$

Higher α_i values indicate items that more sharply differentiate between respondents at different trait levels. Thresholds remain shared across items.

Generalised Partial Credit Model (GPCM)

The GPCM combines item-specific discrimination with item-specific thresholds $\tau_{i,j}$:

$$P(Y_{pi} = c) = \frac{\exp \sum_{j=1}^{c} \alpha_i(\theta_p + \xi_i - \tau_{i,j})}{\sum_{r=1}^{C} \exp \sum_{j=1}^{r} \alpha_i(\theta_p + \xi_i - \tau_{i,j})}$$

Unlike the GRSM, which constrains all items to share the same threshold spacing, the GPCM allows each item to have its own set of threshold parameters. This provides the most flexible parameterisation within the adjacent-category family, but requires estimating substantially more parameters, which can compromise stability with small samples. ### Graded Response Model (GRM)

The GRM uses the cumulative logit link rather than the adjacent-category logit. In its general form (Samejima, 2016), the GRM allows fully item-specific thresholds. The implementation used in this study is a constrained variant in which threshold spacing is shared across items, with item differences captured through a location parameter ξ_i that shifts all thresholds equally. The probability of responding in category c is defined as:

$$P(Y_{pi} = c) = F(\alpha_i(\eta_{pi} - \tau_{c-1})) - F(\alpha_i(\eta_{pi} - \tau_c))$$

where $\eta_{pi} = \theta_p + \xi_i$ is the person-by-item location, $F(x) = \frac{1}{1+\exp(-x)}$ is the logistic CDF, α_i is item discrimination, and the thresholds are ordered: $\tau_1 < \cdots < \tau_{C-1}$, with boundary conditions $\tau_0 = -\infty$ and $\tau_C = +\infty$. The thresholds τ_c are shared population-level parameters; item i's effective thresholds are $\tau_c - \xi_i$, preserving equal inter-threshold spacing across items.

The GRM directly models the cumulative probability $P(Y_{pi} \geq c)$, in contrast to the adjacent-category models which model the odds of each category relative to its neighbour.

Summary of Model Families

Table A1

Summary of ordinal IRT model families used in this study. RSM, GRSM, and GRM share threshold spacing across items; item differences enter only through the location parameter ξ_i. GPCM allows fully item-specific thresholds. The generalised forms (GRSM, GPCM, GRM) estimate item-specific discrimination.

Model	Link	Thresholds	Discrimination
RSM	Adjacent-category	Shared	Equal ($\alpha = 1$)
GRSM	Adjacent-category	Shared	Item-specific
GPCM	Adjacent-category	Item-specific	Item-specific
GRM	Cumulative	Shared (equal spacing)	Item-specific

Dimensional Structures

All models in this study are implemented as generalised linear mixed models (GLMMs) in brms, following the framework described in De Boeck et al. (2011) and Bürkner (2021).

Unidimensional IRT

$$\eta_{pi} = \beta_0 + \xi_i + \theta_p$$

where β_0 is the fixed intercept, $\xi_i \sim \mathcal{N}(0, \sigma^2_{\text{item}})$ is the item-specific random intercept, and $\theta_p \sim \mathcal{N}(0, \sigma^2_\theta)$ is the person-specific random effect.

Correlated-Traits MIRT (CT-MIRT)

Each person p has a vector of dimension-specific traits:

$$\boldsymbol{\theta}_p = (\theta_{pM}, \theta_{pL}, \theta_{pP})^\top \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$$

where $\boldsymbol{\Sigma}$ is a 3×3 covariance matrix. The linear predictor is:

$$\eta_{pi} = \beta_0 + \xi_i + \sum_{d \in \{M,L,P\}} \mathbb{I}_{id}\, \theta_{pd}$$

where $\mathbb{I}_{id} = 1$ if item i belongs to dimension d and 0 otherwise (each item loads on exactly one dimension). The unrestricted covariance matrix $\boldsymbol{\Sigma}$ allows the three latent dimensions to correlate freely, reflecting the possibility that the dimensions are distinct but related constructs.

Bi-Factor MIRT (BF-MIRT)

A general factor θ_p is added alongside dimension-specific factors θ^s_{pd}:

$$\eta_{pi} = \beta_0 + \xi_i + \theta_p + \sum_{d \in \{M,L,P\}} \mathbb{I}_{id}\, \theta^s_{pd}$$

where $\theta_p \sim \mathcal{N}(0, \sigma^2_\theta)$ and $\theta^s_{pd} \sim \mathcal{N}(0, \sigma^2_d)$. The general and specific factors are constrained to be orthogonal:

$$\text{Cov}(\theta_p, \theta^s_{pd}) = 0 \quad \forall\, d$$

This constraint ensures that the specific factors capture only residual variation within each dimension after accounting for the general factor. The distinction from CT-MIRT is structural: the correlated-traits model permits free correlations among dimension-specific traits but includes no general factor, whereas the bi-factor model decomposes variance into a general component shared across all items and orthogonal dimension-specific residuals.

Prior Distributions

Weakly informative priors are used throughout, following recommendations in Bürkner (2021):

Table A2

Prior distributions for all model parameters.

Parameter	Prior	Rationale
Person and item SDs (σ_θ, σ_{item})	$\mathcal{N}(0,3)$	Half-normal (truncated at 0); allows substantial variation while regularising.
Log-discrimination SD	$\mathcal{N}(0,1)$	Covers discrimination values roughly in $[0.14, 7.4]$.
Correlation matrix (CT-MIRT)	LKJ(2)	Mildly favours smaller correlations; $\eta > 1$ concentrates density near the identity matrix.

Appendix B

Instrument items

Teachers were asked to report their endorsement of each statement on a 5-point Likert scale from "strongly disagree" (1) to "strongly agree" (5). Items ending with `_r` are reverse-coded: responses are recoded prior to analysis so that higher scores consistently reflect stronger endorsement of inclusive and equity-oriented beliefs.

Bm: Beliefs About Mathematics (8 Items)

Table B1

Beliefs about mathematics items.

Item	Statement
Bm_1_r	Mathematics is a collection of rules and procedures.
Bm_2_r	Mathematics involves remembering and applying definitions, formulas, mathematical facts and procedures.
Bm_3_r	To do mathematics requires much practice of correct application of routines.
Bm_4_r	When solving mathematical tasks you need to know the correct procedure in advance.
Bm_5	Mathematics involves creativity and new ideas.
Bm_6	Many aspects of mathematics have practical relevance.
Bm_7	Mathematics helps solve everyday problems and tasks.
Bm_8	In mathematics many things can be discovered and tried out by oneself.

Note. Items Bm_1_r through Bm_4_r were removed during the workflow due to poor psychometric properties. The final model was fitted on the remaining 20 items.

Bl: Beliefs About Learners (11 Items)

Table B2
Beliefs about learners items.

Item	Statement
Bl_1_r	Students must be able to solve mathematics problems quickly.
Bl_2	Students must be able to understand why the answer is correct.
Bl_3_r	Students learn mathematics best by attending to the teacher's standard explanations.
Bl_4_r	For students, non-standard procedures can interfere with learning the correct procedure.
Bl_5	Students can figure out a way to solve mathematical problems without a teacher's help.
Bl_6_r	Only the more able students can participate in mathematics activities.
Bl_7_r	To be good at mathematics one needs to be talented.
Bl_8_r	Among other forms of talent, mathematical talent is relatively fixed.
Bl_9_r	Boys tend to be more talented in mathematics than girls.
Bl_10_r	Attention to mathematically weak students can interfere with the learning experience of other students.
Bl_11_r	Attention to mathematically talented students can interfere with the learning experience of other students.

Bp: Beliefs About Pedagogy (5 Items)

Table B3
Beliefs about pedagogy items.

Item	Statement
Bp_1_r	Innovative pedagogy (e.g., project-based learning, flipped classroom) is not worth the time and expense.
Bp_2_r	Innovative pedagogy (e.g., project-based learning, flipped classroom) should only play a minor, supplementary role in the mathematics classroom.
Bp_3	I'm continuously seeking better ways to teach mathematics.
Bp_4	Teachers should help students using mathematics to critically analyse the world.
Bp_5	Discussion of social issues should be involved in the maths classroom.